

Vocal repertoire, individual acoustic distinctiveness, and social networks in a group of captive Asian small-clawed otters (*Aonyx cinerea*)

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Communication in solitary and social mustelids has been mainly studied in terms of olfactory abilities. However, most otters appear highly social and vocal, but little is known about the role of vocal signals in communication in those species. An analysis of the airborne vocal repertoire and social networks of Asian small-clawed otters (*Aonyx cinerea*) is reported herein. The vocal repertoire of a captive group of 10 individuals was composed of 4 context-dependent vocalization units, which were emitted alone, repeated, or combined in a nonrandom way, resulting in 7 vocalization types. Individuals varied in their preference to use certain vocalization types and to maintain bonds with some particular group members. Older, larger, and heavier otters produced lower-frequency sounds. Thus, Asian small-clawed otters are clearly social and possess a high potential for vocal communication, such as acoustic coding of messages about ongoing social activities and biologically important characteristics about the sender. Future comparative studies of mustelids will tell whether solitary species may use essentially olfactory communication, whereas social species use multimodal communication, including vocal cues.

Key words: *Aonyx cinerea*, Asian small clawed otter, mustelid, otter, social network, vocalization, vocal repertoire

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The question of coevolution between vocal communication and social systems, notably in mammals, is the subject of lively discussions (Lemasson 2011). Although the structure of the vocal repertoire is strongly linked to phylogeny (Cap et al. 2008), there is some evidence that social structure might also have significant influence upon the communication system (Lemasson 2011). Studies have found correlations between social factors (group size, strength of bonds, and level of mobility) and vocal parameters (repertoire size and level of acoustic variability) in a few animal taxa (ground-dwelling sciurid rodents [22 species—Blumstein and Armitage 1997], Carolina chickadees [*Poecile carolinensis*—Freeberg 2006], silvery mole-rats [*Heliophobius argenteocinerea*—Knotková et al. 2009], whales [64 species—May-Collado et al. 2007], and nonhuman primates [42 species—McComb and Semple 2005]). One way to contribute to a better understanding of social–vocal emergence is to study animal taxa from phylogenetically related species, which have various social systems.

Mustelids exhibit a large diversity in their social life and some species are said to be more social than others (Johnson et

al. 2000; Lodé et al. 2003). For example, otters display considerable variations in their social structure from a solitary life to the monogamous pair, or to the extended family group, and sometimes with bachelor groups of males. Because at one time a strong focus was put on functions related to territory protection and mate attraction, what is known about the communication system of mustelids concerns primarily individualistic habits and intergroup communication associated to olfactory cues (Eisenberg and Kleiman 1972; Powell 1979; Lodé et al. 2003). Communication through scent marks, which allows the discrimination of sex, dominance, and familiarity, is well known in mustelids (e.g., domestic ferrets [*Mustela putorius furo*—Clapperton et al. 1988]) and other solitary carnivores (e.g., brown bears [*Ursus arctos*—Clapham et al. 2012]). However, little is known about vocal behavior of mustelids because the topic has only been studied in a few species (e.g., European badgers [*Meles meles*—Wong et al.



1999] and sea otters [*Enhydra lutris*—McShane et al. 1995]). The social–vocal coevolution theory predicts that a more socially complex life is associated with a more complex vocal repertoire (Lemasson 2011). Hence, vocalizations also were expected to play a key role in a group-living mustelid species, notably to regulate daily intragroup activities and social interactions (Wong et al. 1999). Interestingly, in the yellow mongoose (*Cynictis penicillata*), another carnivorous mammal, the vocal repertoire of facultatively social species is larger than that of solitary species, but smaller and less context-specific than in obligate social species (Le Roux et al. 2009).

Mammal vocalizations usually carry 3 different kinds of information: referential, emotional, and the sender's identity (e.g., see the 3 kinds of information encoded in the vocalizations of Campbell's monkeys [*Cercopithecus campbelli*]: type of danger spotted [Ouattara et al. 2009]; arousal state of sender [Lemasson et al. 2012]; and age, sex, and identity of the caller [Lemasson and Hausberger 2011]). Among those types of information, the acoustic coding of an individual's identity is of primary interest because it functions to maintain spatial and social cohesion between group members. Individual acoustic variations in mammals can be due to differences in dominance status (e.g., yellow baboons [*Papio cynocephalus*—Fischer et al. 2004], red deer [*Cervus elaphus*—Fitch and Reby 2001], spotted hyenas [*Crocuta crocuta*—Mathevon et al. 2010], and domestic horses [*Equus caballus*—Rubenstein and Hack 1992]), or social affinities (bottlenose dolphins [*Tursiops truncatus*—Janik 2000] and Campbell's monkeys [Lemasson and Hausberger 2011]), as well as to individual morphological differences (e.g., body size impacting the vocal apparatus: nonhuman primates [12 species—Ey et al. 2007], killer whales [*Orcinus orca*—Kremers et al. 2012], and southern elephant seals [*Mirounga leonina*—Sanvito et al. 2007b]).

Asian small-clawed otters (*Aonyx cinerea*) are considered social and vocal among mustelids with diurnal habits (Timmis 1971; Furuya 1976; Sivasothi and Nor 1994; Johnson et al. 2000; Larivière 2003), although little is known about their intragroup social networks and vocal communicative abilities. They live in humid areas of Southeast Asia, such as rivers and swamps, feeding primarily on fish, crabs, and invertebrates (Kruuk et al. 1994), and contrary to other otter species, they spend the majority of their lives in terrestrial habitats (Larivière 2003). They have a good visual acuity both underwater and aerially (Balliet and Schusterman 1971). They form extended family groups, a monogamous pair with some young adults and cubs (6–12 individuals), and both parents seem to participate in the care of offspring (Leslie 1970; Furuya 1976; Foster-Turley 1986; Foster-Turley and Engfer 1988). However, the species is socially very flexible in a variable environment and sometimes males gather in small bachelor groups (Furuya 1976). Dispersal of young is very flexible. They can stay with parents for several years or can disperse when the next litter is born. Gestation is about 60 days with 1 (rarely 2) litter of 1–6 young (2 or 3 on average) per year (reviewed by Larivière 2003). Sexual

TABLE 1.—Individual characteristics of captive Asian small-clawed otters (*Aonyx cinerea*). Fa: father; Mo: mother; others are offspring (letters represent 3 litters). M: male; F: female.

Individual identification	Age (years)	Sex	Weight (kg)	Body length (cm, without tail)
Fa	13	M	3.47	43
Mo	10	F	3.2	40
A1	2.5	F	3.34	40
A2	2.5	F	2.84	39
A3	2.5	F	3.25	39
B1	1.5	F	2.29	37
B2	1.5	F	3.24	39
C1	0.5	F	1.55	35
C2	0.5	M	2.13	34
C3	0.5	F	1.67	35

maturity likely occurs during the 1st year and longevity may reach 15 years in captivity (Larivière 2003).

Observations and acoustic recordings were conducted on a captive group of Asian small-clawed otters at Planete Sauvage Animal Park in Port-Saint-Père, France, to assess their level of sociality and vocal communication abilities. First, the vocal repertoire of the group was defined by classifying acoustic structures into vocalization types and by identifying an individual's preferred vocalization type. The vocal repertoire of this social species was expected to be composed of a high number of acoustically distinguishable vocalization types with the call rate depending on the social status of the sender (Bouchet et al. 2012). Second, information potentially carried by vocalizations was examined. Vocalization types used by otters were expected to vary during different social contexts. Third, acoustic structures of contact vocalizations were expected to carry reliable information about caller's identity and morphology (Lemasson and Hausberger 2011). Fourth, social networks were analyzed by testing how and with whom the different group members interacted.

MATERIALS AND METHODS

Study group.—The group of Asian small-clawed otters studied was a family composed of 10 captive-born individuals (Table 1): 1 adult pair (Fa: father, Mo: mother) and 3 sequential litters of their offspring, born 1 year apart, matching the social structure described in the wild (Furuya 1976). Siblings were differentiated by litter number (A1–A3, B1 and B2, and C1–C3). They were housed at Planete Sauvage Animal Park (Port-Saint-Père, France) in an indoor (1 × 0.8 × 1.32 m, with 2 levels) and an outdoor (240 m²) enclosure. The outdoor part was enriched with a pond (diameter: 4 m, depth: 1 m), a small waterfall, and several rocks. The indoor part was enriched with straw litter. The group was fed 4 times a day, with dead chicks (0830 h) and fish (1100, 1430, and 1700 h). Water was available ad libitum.

Protocol and analyses.—Data were collected from April to June 2011. The first 3 weeks were used for the otters to habituate to the presence of the unfamiliar human observer (M-AM) carrying a recording apparatus, and to a temporary mark

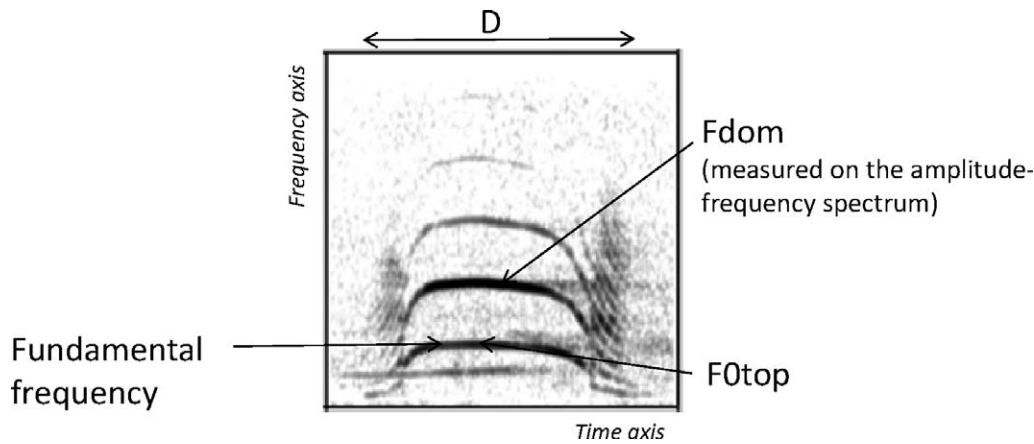


FIG. 1.—Illustration of the acoustic measurements performed on the vocalizations of captive Asian small-clawed otters (*Aonyx cinerea*; accuracy: 94 Hz/1 ms).

using shaved rings on the tails to facilitate the recognition of individuals.

Data were collected using 3 standard sampling methods in ethology (Altmann 1974), covering all time periods from 0830 to 1830 h. First, the vocalizations were recorded, as well as the immediate context of emission (see Appendix I for definitions)

using all-occurrence sampling performed at the group level. During 32.5 h of recordings, a total of 1,093 vocalizations were uttered, for which the identity of the caller was known. In a few cases ($n = 36/1,093$), the associated context of emission could not be identified. Recordings were done using a PMD660 digital recorder (sampling rate: 48 kHz; amplitude resolution:

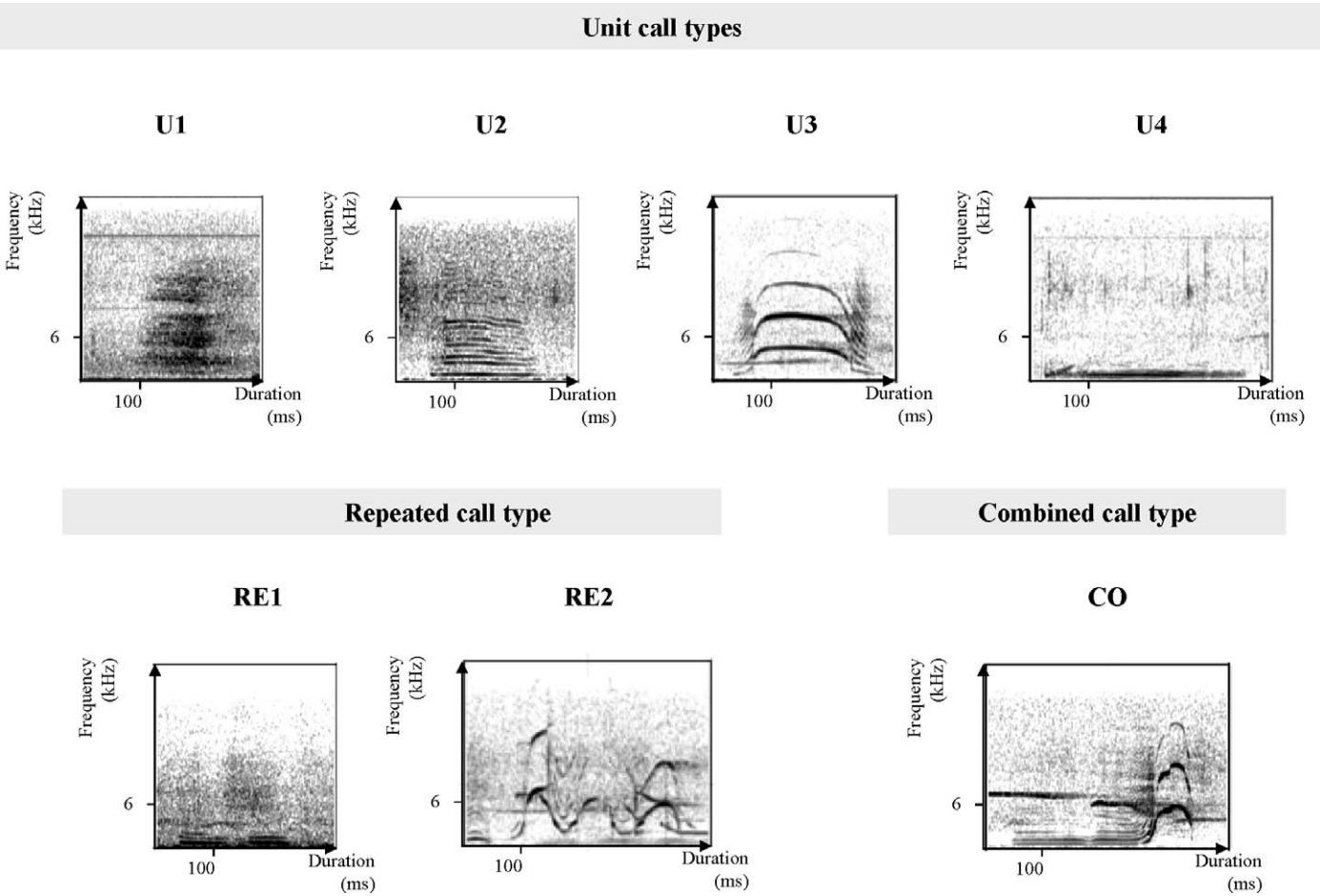


FIG. 2.—Classification of vocalization types of captive Asian small-clawed otters (*Aonyx cinerea*).

TABLE 2.—Acoustic characteristics of vocalization types of captive Asian small-clawed otters (*Aonyx cinerea*). A) Acoustic measurements performed on $n = 30$ vocalization exemplars for each vocalization type: D = duration, F0top = highest point of the fundamental frequency, Fdom = dominant frequency (each cell: $\bar{X} \pm SE$). B) Discriminant function (DF) analysis testing the validity of the classification into 4 vocalizations.

	U1	U2	U3	U4	DF1	DF2
A)						
Measurements						
D (ms)	198 $\pm$ 9	303 $\pm$ 45	137 $\pm$ 10	2,479 $\pm$ 234		
F0top (Hz)	3,217 $\pm$ 156	2,423 $\pm$ 393	5,221 $\pm$ 430	951 $\pm$ 49		
Fdom (Hz)	2,613 $\pm$ 184	2,519 $\pm$ 376	4,450 $\pm$ 496	276 $\pm$ 25		
B)						
Discriminant function						
Canonical correlation coefficient					0.870	0.529
Wilks' lambda					0.169	0.696
P-value					< 0.0001	< 0.0001
Standardized canonical correlation coefficient					0.87	−1.30
Correlated variable (coefficient of correlation)					D (0.86)	F0top (−0.82)

16 bits; Marantz, Eindhoven, Netherlands) connected to 1 K6/ME66 directional microphone (Sennheiser Communication, Ivry sur seine, France; to record animal vocalizations) and 1 ECM-T6 tie-microphone (Sony, Puteaux, France; to record observer's comments).

In the laboratory, several acoustic measurements (Fig. 1) were taken from real-time spectrograms (D: duration, Fdom: dominant frequency [i.e., frequency presenting the highest amplitude], and F0top: highest point of the fundamental frequency). Spectrograms were displayed using Avisoft software (SASLab [Specht 1996], Fast Fourier Transform 512, at a temporal resolution of 87.5%). These data were used to preclassify vocalizations into potential types, based on a visual inspection of spectrograms and listening (as in Lemasson and Hausberger [2011]), confirm the classification using acoustic measurements (based on a standard discriminant function analysis, as in Kremers et al. [2012], performed on a subset of $n = 30$ vocalization exemplars randomly selected for each preclassified vocalization type and emitted by at least 5 different otters per type), test for individual preference to emit certain vocalization types (using *G*-tests), analyze the main context of each vocalization type (using *G*-tests as in Bouchet et al. [2012]), test for the existence of individually distinctive acoustic structures (using Kruskal–Wallis tests and standard discriminant function analysis), and examine possible correlations between the sender's age and acoustic parameters (using Spearman's correlation tests).

Second, all-occurrence sampling at the individual level was performed to count the exact number of vocalizations per 10-min period emitted by each group member ($n = 5$ per otter). Differences between individual vocalization rates were tested using *G*-tests.

Third, individual focal sampling (33 sessions of 5 min per individual) was performed to quantify nonvocal interactions (social resting: sleeping or resting in physical contact with another group member; social playing: play-chase and play-fight; affiliative interaction: allogrooming, initiation and maintenance of spatial proximity, and mouth-to-mouth contact; and aggressive interaction: biting, chasing, and pushing) with

the different group members. Those data were used to draw sociograms based on the analysis of partner preferences (using *G*-tests) and to test for mother versus father and same-age versus different-age sibling preferences (using Wilcoxon signed-rank and Mann–Whitney *U*-tests, respectively).

Because acoustic variables were not normally distributed and did not have equal variances, nonparametric statistics were used. Statistical analyses were performed using SPSS 16.0 (SPSS Inc. 2007—for discriminant function analysis), Microsoft Excel (Microsoft 2007—for *G*-tests), and Statistica 8.0 (StataCorp 2007—for other tests) software. All tests were 2-tailed with significance set at $P = 0.05$. For each *G*-test and Spearman test, a post hoc analysis of statistical power was calculated using Gpower 3.1 (Buchner et al. 2009). All statistical tests presented below had sufficient power to reject or accept the null hypothesis at $1 - \beta > 0.95$.

This research was only observational and followed guidelines of the American Society of Mammalogists (Sikes et al. 2011). Animal care and research protocols used in this work complied with the current French laws governing animal research and were approved by the Direction Départementale des Services Vétérinaires committee of Loire Atlantique prefecture.

RESULTS

Classification of the vocal repertoire based on acoustic structure.—The 1,093 recorded vocalizations were classified into 7 vocalizations used in 3 broad categories: single vocalization, repeated vocalizations, and combined vocalizations. Four vocalization types were identified, as follows: U1 (atonal), U2 (many harmonics and no frequency modulation), U3 (many harmonics with an arched frequency modulation), and U4 (no harmonics or frequency modulation). Two repeated vocalizations were classified as RE1 (repeats of U2) and RE2 (repeats of U3). Combined vocalizations (CO) consisted of calls composed of a U2 and a U3 unit (Fig. 2).

Vocalizations types differed significantly from one another in their acoustic structure (Table 2A). A discriminant function analysis based on 3 acoustic measurements (D, F0top, and

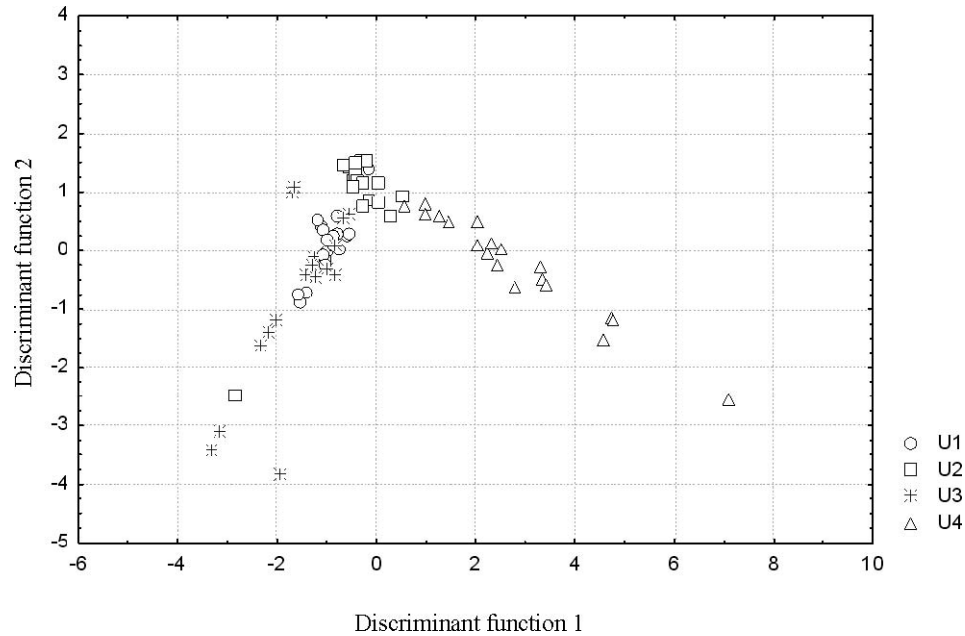


FIG. 3.—Acoustic distinctiveness between vocalization types of captive Asian small-clawed otters (*Aonyx cinerea*): scatter plot of the discriminant function analysis. Each small symbol represents a unit vocalization type.

Fdom) confirmed the pertinence of the classification into 4 vocalization types (Table 2B; Fig. 3). This discriminant function analysis was done using half of the measured vocalizations (i.e., $n = 15$ per vocalization type, randomly selected) as the training sample and the other half as the test sample. The percentage of correctly classified vocalizations was much higher than expected by chance (64%, Wilks' lambda = 0.169, $F = 19.21$, $P < 0.0001$). Discriminant functions 1 and 2 were, respectively, highly correlated to the vocalization duration (D) and to the fundamental frequency (F0top [Table 2B]).

However, important acoustic variations within some vocalization types were found (Fig. 4; Table 2A). U2 vocalizations were graded in duration. U3 vocalizations presented a more or less complete arched frequency modulation. RE2 vocalizations were composed of different arches merged to varying degrees. CO vocalizations combined U3 with complete or incomplete arches, optionally repeated, with U2. U2 vocalizations were placed before or after U3.

Individual preferences and vocalization rates.— All vocalization types were produced by parents and offspring of all age cohorts (Table 3). This was a prerequisite to consider them as specific vocalization types and not an individual signature. Although the father ($G = 1,670$, $P < 0.001$) and offspring ($410 < G < 63$, $P < 0.001$; not testable for 3 offspring) emitted primarily U3 vocalizations, the mother emitted primarily U1 and U4 vocalizations ($G = 159$, $P < 0.001$).

Vocalization rates also differed among individual otters ($G = 10$, $P < 0.05$), with the father and 1 offspring (B2) showing the highest rates, and the lowest rate was by 3 offspring (A2, C2, and C3); the former otter vocalized greater than 20 times more often than the latter.

Contexts of emission.—Vocalization types differed from one another with regard to their most important contexts of emission, except for RE1 and RE2, which were not emitted frequently enough to be tested (Table 4; Appendix I). All other vocalization types were characterized by 1–3 most important contexts of emission. Moreover, the predominant context of the combined CO vocalization type also was the most important context of the constituting unit vocalization type (i.e., U2, $G = 102$, $P < 0.001$).

Concerning the contexts of using the 4 vocalization types: U1 was primarily produced during exploration and vigilance (e.g., looking at group members or at the observer or outside the enclosure [$G = 125$, $P < 0.001$]). U2 was primarily associated to food-related behaviors (e.g., beg for food [$G = 37$, $P < 0.001$]). U3 was produced in a very large range of contexts including spatial isolation, exploration, and sociopositive interactions (e.g., walking alone, staying far from others, looking outside the enclosure, social gaze, approach, play, and group travel [$G = 680$, $P < 0.001$]). U4 was primarily produced during food-related interactions (e.g., food offering or protection [$G = 123$, $P < 0.001$]).

Identity coding.—The acoustic structure of U3 vocalizations, which were emitted with the highest rates and in a large variety of contexts, varied significantly between individuals for all 3 measured acoustic parameters (Table 5): duration (Kruskal–Wallis tests, $H_{7,88} = 19.32$, $P < 0.01$), fundamental frequency ($H_{7,88} = 61.63$, $P < 0.001$), and dominant frequency ($H_{7,88} = 61.87$, $P < 0.001$). A discriminant function analysis confirmed that a given vocalization could reliably be associated to the appropriate sender (percentage of correctly classified vocalizations: 51.1%, Wilks' lambda = 0.119, $F_{28} = 8.018$, $P < 0.001$).

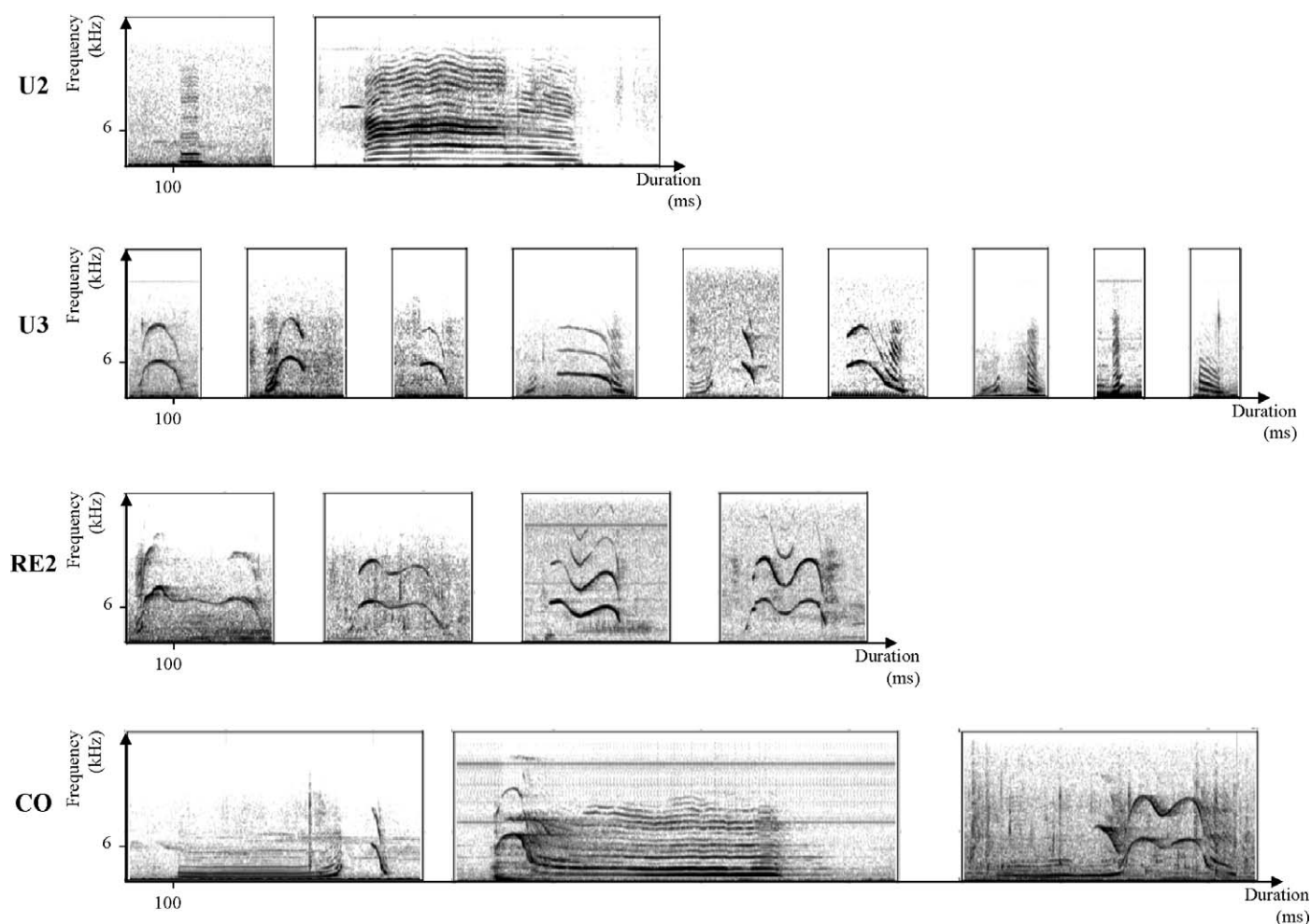


FIG. 4.—Acoustic variations within some vocalization types of captive Asian small-clawed otters (*Aonyx cinerea*).

Upon comparison, the otter's age, body length, and body mass were negatively correlated with frequency-related parameters, fundamental frequency (Spearman tests, $n = 8$; age: $r = -0.78$, $P < 0.05$; body mass: $r = -0.76$, $P < 0.05$; and body length: $r = -0.81$, $P < 0.05$), and dominant frequency ($n = 8$; age: $r = -0.84$, $P < 0.01$; body mass: $r = -0.76$, $P < 0.05$;

and body length: $r = -0.88$, $P < 0.01$), but not with vocalization duration ($n = 8$; age: $r = -0.062$, $P = 0.88$; body mass: $r = -0.071$, $P = 0.87$; and body length: $r = -0.073$, $P = 0.86$).

Social networks.— Among the 4 behavioral categories investigated, the mean events per minute per individual $\pm SE$

TABLE 3.—Individual preferences of captive Asian small-clawed otters (*Aonyx cinerea*) in production of different vocalization types and their vocalization rate. Upper cells: number of vocalizations recorded per vocalization type and per individual. G -value/ P -value: results of the G -tests done for each individual (boldface numbers indicate contexts highly contributing to the G -analysis with partial G -value ≥ 10). NT: not testable due to low sample size. Last row: vocalization rate per minute per individual. Fa: father; Mo: mother; others are offspring (letters represent 3 litters). Vocalization types are defined in the “Results.”

Vocalization type	Fa	Mo	A1	A2	A3	B1	B2	C1	C2	C3
U1	0	36	2	7	6	5	18	2	0	0
U2	4	4	0	0	10	6	10	7	0	2
U3	474	10	13	23	37	70	38	129	15	2
U4	8	51	0	3	7	3	5	0	0	0
RE1	2	4	0	0	0	0	2	1	0	0
RE2	3	0	7	2	2	0	3	0	0	0
CO	8	0	3	1	11	8	4	21	0	4
G -value	1,670	159	NT	63	75	198	73	410	NT	NT
P -value	< 0.001	< 0.001		< 0.001	< 0.001	< 0.001	< 0.001	< 0.001		
Vocalization rate	2.96	0.74	0.40	0.20	0.32	0.50	0.60	2.72	0.12	0.00

TABLE 4.—General contextual profile of use by captive Asian small-clawed otters (*Aonyx cinerea*) of each vocalization type (defined in the “Results”). General contexts of emission (see detailed values in Appendix I) *G*-tests were done for each vocalization type (results are presented in the 2 last rows of the table); boldface numbers indicate contexts highly contributing to the *G*-analysis (partial *G*-value ≥ 10). NT: not testable due to low sample size.

Context	U1 (<i>n</i> = 76)	U2 (<i>n</i> = 43)	U3 (<i>n</i> = 811)	U4 (<i>n</i> = 77)	RE1 (<i>n</i> = 9)	RE2 (<i>n</i> = 17)	CO (<i>n</i> = 60)
Social isolation	15	4	229	0	3	0	1
Exploration	44	13	262	13	2	5	9
Sociopositive	13	7	187	9	1	9	10
Food	0	15	28	47	1	0	37
Self-directed	1	1	78	2	1	0	3
Socionegative	0	3	3	0	1	0	0
Not identified	3	0	24	6	0	3	0
<i>G</i> -value	125	37	680	123	NT	NT	102
<i>P</i> -value	< 0.001	< 0.001	< 0.001	< 0.001			< 0.001

was 0.244 ± 0.022 , almost twice as frequent as social resting (0.156 ± 0.027), and social play (0.114 ± 0.025) initiations were 13 times more frequent than aggressive interactions (0.019 ± 0.006).

The parents had a preferential bond, and frequently interacted affiliatively with one another, slept together, and never interacted aggressively (Fig. 5). Offspring initiated more affiliative interactions with their father than with their mother (Wilcoxon test, $n = 8$, $T = 0$, $Z = 2.02$, $P < 0.05$). The mother was a preferred target for social rest ($n = 8$, $T = 0$, $Z = 2.37$, $P < 0.05$), particularly in the youngest offspring (Fig. 5).

Among offspring, resting partners were preferentially selected among juveniles of the same age (e.g., C1-C2, C2-C3, C3-C1, and A2-A3 [Mann-Whitney *U*-tests, $U = 11.5$, $n_1 = 7$, $n_2 = 21$, $P < 0.001$; Fig. 5]). No age bias was found in the networks of social play ($U = 51$, $P = 0.249$) because they often involved juveniles of different ages (e.g., A1-B2, A2-B2, A2-B1, and A3-B1 [Fig. 5]). Again, no age bias was found in affiliative ($U = 67$, $P = 0.756$) and aggressive ($U = 64.5$, $P = 0.64$) interactions. To a lesser extent, the father also was targeted as a play partner (Fig. 5). Only 1 pair of individuals appeared frequently involved in aggressive interactions (C2-C1 [Fig. 5]).

DISCUSSION

It must be acknowledged that the results discussed herein are based on the study of a single captive family group. Extrapolation about social and vocal behavior in wild populations of this otter species requires more comparative studies. This captive group of Asian small-clawed otters produced a relatively complex vocal repertoire and vocaliza-

tion types carried potential information concerning context and the identity and morphology of the individual otter. Vocalization rates and social networks revealed a different social role for each of the 2 parents, as well as different social bonds among siblings of the same versus different ages.

Four structurally differentiable vocalizations were identified that could be emitted alone, repeated, or combined in a nonrandom way, resulting in a vocal repertoire composed of 7 vocalization types. Also, given the high level of gradation observed in some vocalization types, the vocal system under study could potentially be even more complex. The size of the vocal repertoire is within the range previously suspected in other social otter species, as well as pinniped species (number of vocalization types identified in sea otters was 10 [McShane et al. 1995]; and in Weddell seals [*Leptonychotes weddellii*], leopard seals [*Hydrurga leptonyx*], and Ross seals [*Ommatophoca rossii*], respectively, was 14, 7, and 5 [van Opzeeland et al. 2010]). However, different acoustical studies are not easily comparable because repertoire size directly depends on the criteria used to classify vocalizations. For example, using definitions comparable with ours, Thomas and Kuechle (1982) classified vocalizations of Weddell seals into 12 categories, subdivisible into more than 34 types. Another difficulty is that studies do not always describe any attributes of the putative vocalization types, making the vocal repertoire size somewhat arbitrary (e.g., Timmis 1971). The contextual analysis confirms our classification into 4 vocalizations. However, the immediate contexts of the repeated and combined vocalization types did not differ fundamentally from the contexts of the single vocalization types. In some bird (Freeberg and Lucas 2002) and primate (Ouattara et al. 2009; Lemasson et al. 2010) species sound repetition and combination carry different

TABLE 5.—Acoustic measurements on vocalization type U3 ($n = 11$) from captive Asian small-clawed otters (*Aonyx cinerea*): D = duration, F0top = highest point of the fundamental frequency, Fdom = dominant frequency (each cell: $\bar{X} \pm SE$). The mother (Mo) could not be included in the analysis because her U3 arches were all too incomplete. C3 was not included in this analysis because of the limited number of vocalizations recorded from her. Fa: father; others are offspring (letters represent 3 litters).

Measurements	Fa	A1	A2	A3	B1	B2	C1	C2
D (ms)	103 $\pm$ 11	91 $\pm$ 20	84 $\pm$ 11	94 $\pm$ 9	69 $\pm$ 6	72 $\pm$ 7	138 $\pm$ 17	102 $\pm$ 8
F0top (Hz)	4,860 $\pm$ 343	5,819 $\pm$ 408	6,600 $\pm$ 234	7,412 $\pm$ 337	6,590 $\pm$ 234	7,258 $\pm$ 227	10,714 $\pm$ 355	10,174 $\pm$ 490
Fdom (Hz)	4,424 $\pm$ 349	5,311 $\pm$ 384	6,144 $\pm$ 200	6,857 $\pm$ 402	6,203 $\pm$ 216	6,827 $\pm$ 202	9,635 $\pm$ 359	9,794 $\pm$ 536

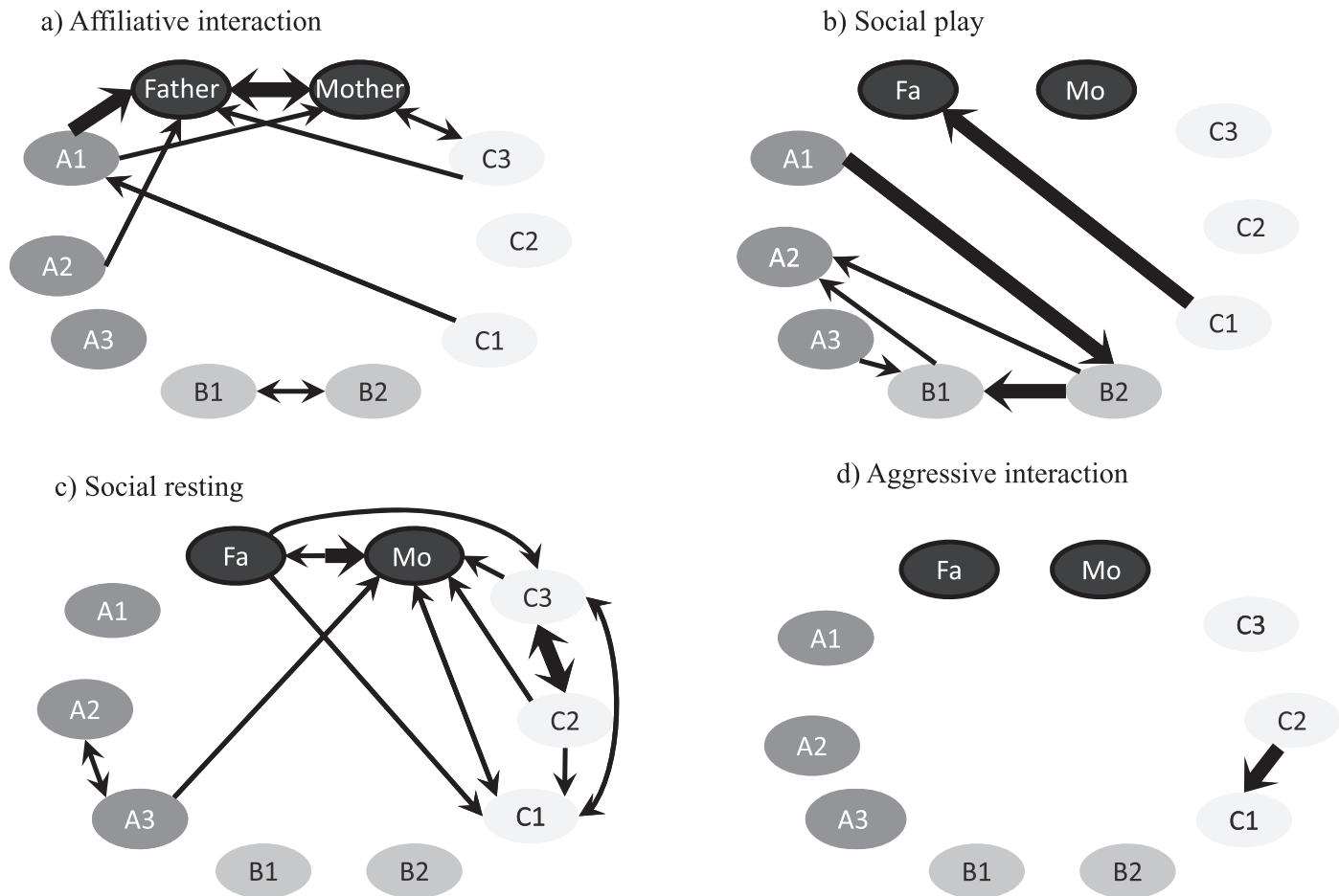


FIG. 5.—Sociograms of captive Asian small-clawed otters (*Aonyx cinerea*) based on a) affiliative interactions, b) social play, c) social resting, and d) aggressive interactions. The arrows are directed from the emitter to the receiver of a behavior. Only interactions occurring more often than expected by chance for a given emitter are represented (*G*-tests per individual, $P < 0.05$). The arrow's thickness is correlated to the partial *G*-value (thin arrows: $5 < \text{partial } G < 10$; thick arrows: $\text{partial } G \geq 10$).

information than the constituting sound units. In the case of this study, vocalization combination, repetition, and gradation can be the result of different factors, such as subtle contextual differences not identified with our protocol, individual particularities, as well as different levels of emotional state or maturity of the otter.

Only hypotheses can be raised at this stage regarding the social functions of the 4 vocalization types in the Asian small-clawed otter. Given the context associated with each vocalization, it is likely that U1 functions as an alarm vocalization, U2 and U4 as food-related vocalizations (U2: food request, and U4: food possession), and U3 as a contact vocalization. This last category usually corresponds to the most frequently uttered vocalization type, which is emitted all day long and used to maintain the cohesiveness of the group and to coordinate group movements. Those 3 vocal functions are widespread in the animal kingdom (e.g., various mammals and birds—Manser et al. 2001; Hollen and Radford 2009; Clay et al. 2012).

Contact vocalizations, more than other vocalization types, usually carry reliable information to identify the caller

(Lemasson and Hausberger 2011). It was therefore predictable that U3 acoustic structures would be individually distinctive; this is particularly useful to locate group members in a relatively visually limited environment, such as these otter's preferred habitat (Sivasothi and Nor 1994). Individually distinctive acoustic structures have been demonstrated in a large number of mammals, for example, white-nosed coatis (*Nasua narica*—Maurello et al. 2000), fallow deer (*Dama dama*—Vannoni and McElligott 2007), bottlenose dolphins (Janik 2000), dogs (*Canis lupus familiaris*—Yin and McCowan 2004), and Campbell's monkeys (Lemasson and Hausberger 2011), including all pinniped species (reviewed by Insley et al. 2003). This system of acoustic signatures enables mammals to discriminate the voices of social group members (e.g., sheep [*Ovis aries*—Searby and Jouventin 2003], fur seals [*Arctocephalus tropicalis*—Charrier et al. 2003], suricates [*Suricata suricatta*—Townsend et al. 2012], olive baboons [*Papio anubis*—Lemasson et al. 2008], and pinniped species [Insley et al. 2003]).

In this study, individual acoustic distinctiveness was related to individual morphological distinctiveness. Correlations

between acoustic parameters and morphology or age have been shown in domestic horses (Lemasson et al. 2009), nonhuman primates (comparison of 12 species—Ey et al. 2007), killer whales (Kremers et al. 2012); koalas (*Phascolarctos cinerea*—Charlton et al. 2011), and sea otters (McShane et al. 1995). In most mammals, the anatomy of the vocal apparatus, correlated to the animal's body size, determines the acoustic features of its voice (larger lungs predicts longer vocalizations, longer and thicker vocal folds predict lower fundamental frequency, and longer vocal tract predicts formants with energy concentrated in lower frequencies [Ey et al. 2007]). Here, larger Asian small-clawed otters did not produce longer vocalizations, but produced lower fundamental and dominant frequencies. Significant correlations were found between vocal tract length, body mass, and formant dispersion in dogs and southern elephant seals, confirming that energy dispersion can deliver information about sender's body size (Riede and Fitch 1999; Sanvito et al. 2007b). In this case, this would allow otters to assess the age and size of a newcomer detected acoustically at the territorial border and could play a role in territory protection and mate attraction, such as in red deer, in which females prefer the roars of larger males (Charlton et al. 2007). Experiments are now needed to confirm any auditory decoding ability in more species.

Social networks of the Asian small-clawed otter appear as complex as vocal repertoires, with notably nonrandom preferential bonds. In general, interactions were more positive than negative as also was found in North American river otters (*Lontra canadensis*—Hansen et al. 2009). The 2 parents had a privileged bond, but did not present the same social status for their offspring. The father was a preferred target for affiliative interactions. The father also played a key role in intragroup cohesion, preferentially emitting contact vocalizations and showing the highest vocal activity in the group. In contrast, the mother was preferred for social rest. She also seemed responsible for environmental monitoring and emitted high rates of alarm vocalizations. Among the offspring, individuals tended to interact preferentially with same-age siblings. Agonism was only directed from the single immature male toward an immature female sibling. It must be acknowledged that the aforementioned conclusions are based on the social description of a single captive group with only 1 father (i.e., only 1 mature male) and 1 mother, and with an unbalanced sex ratio.

Asian small-clawed otters exhibit a rich vocal repertoire. Moreover, numerous features, such as context-dependent vocalization types and vocalization concatenation reveal the importance of vocalizations within this group-living mustelid species, conversely to solitary species that rely essentially on olfaction to communicate (Eisenberg and Kleiman 1972; Powell 1979; Clapperton et al. 1988; Lodé et al. 2003). Vocal signals could enhance social interactions, particularly by providing information about the otter's identity and its ongoing activity, opening interesting perspectives for future experimental research. Comparative studies with more species still need to test whether multimodal communication, notably mixing

olfactory and vocal cues, emerged in parallel with sociality among mustelids. Concerning Asian small-clawed otters, further experimental studies using notably fake predators and food provisioning are now necessary to confirm the functions of the different vocalization types. Playback experiments also are needed to test the decoding abilities of animals. In-air audiograms in 2 related species have shown that the variations in acoustic pattern detected in our work falls into their range of hearing (North American river otter: 460–33,000 Hz [Gunn 1988]; and southern sea otter: 125–32,000 Hz [Ghoul and Reichmuth 2012]). In a recent experiment, Lemasson et al. (2013) found that female Asian small-clawed otters discriminated between familiar and unfamiliar adult males based on their sound (playback of U3 calls) and the smell of feces. Work on other marine carnivores, such as northern elephant seals (*Mirounga angustirostris*), showed that decoding abilities go beyond simple individual recognition, for example, vocal signal directionality (Holt et al. 2010). To confirm the particular social status of the parents and the importance of offspring age, comparative observations on more groups of Asian small-clawed otters are needed, notably in the wild. Finally, whether or not vocal learning from social group members occurs in this species, as it occurs in pinnipeds (Sanvito et al. 2007a; Schusterman 2008), remains an open question.

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APPENDIX I

Detailed contextual profile of the usage of each vocalization type by captive Asian small-clawed otters (*Aonyx cinerea*). Vocalization types are defined in the “Results.”

Context ^a	U1 (n = 76)	U2 (n = 43)	U3 (n = 811)	U4 (n = 77)	RE1 (n = 9)	RE2 (n = 17)	CO (n = 60)
1	1	4	97	0	3	0	1
2	9	7	138	0	0	2	6
3	23	0	51	13	0	0	1
4	0	8	24	0	1	0	35
5	0	0	114	0	0	0	0
6	1	3	43	0	2	3	2
7	0	0	0	28	0	0	0
8	10	3	15	0	0	0	0
9	0	0	5	0	0	0	0
10	9	1	44	9	0	6	3
11	4	2	35	0	0	3	0
12	14	0	18	0	0	0	0
13	0	0	43	0	0	0	1
14	0	1	1	19	0	0	0
15	0	0	89	0	0	0	1
16	1	1	12	1	1	0	2
17	0	0	17	1	0	0	0
18	0	0	10	0	0	0	0
19	0	0	4	0	0	0	0
20	0	0	2	0	0	0	0
21	0	1	1	0	0	0	2
22	1	0	0	0	0	0	0
23	0	0	9	0	0	0	0
24	0	4	0	0	0	0	6
25	0	3	2	0	1	0	0
26	0	0	1	0	0	0	0
27	0	4	0	0	0	0	0
28	0	0	10	0	1	0	0
29	0	1	2	0	0	0	0
30	3	0	24	6	0	3	0

^a 1: Walk alone at more than 1 m from others; 2: look outside the enclosure (> 1 s); 3: look at a group member (> 1 s); 4: beg for food by leaning toward a group member manipulating a food item; 5: stay immobile and spatially isolated at more than 5 m from others; 6: approach or interact with the observer; 7: offer food to another group member by dropping a piece of fish in front of a group member; 8: look at the observer (> 1 s); 9: look inside the indoor shelter (> 1 s); 10: social play; 11: approach a group member; 12: approach a spatially isolated (> 5 m) group member; 13: rub back on the ground or a wall; 14: prevent a group member from taking its piece of fish by keeping it in its mouth or paws; 15: travel with all individuals in the group maintaining a spatial cohesion; 16: self-grooming; 17: rest; 18: explore the enclosure with the legs and nose; 19: self-scratching; 20: solitary play; 21: having a piece of fish stolen by another group member; 22: aggression toward the observer; 23: follow a group member; 24: rub against a group member; 25: targeted with aggression by another group member; 26: display aggression toward another group member; 27: eat; 28: allogrooming; 29: steal a piece of fish from a group member; and 30: context not identified.